

Analysis of Algorithms

CPS 3440* 02, CPS 3440* 03, and CPS 5014* 01

Dr. Yan Ma

yama@kean.edu

- Before we start, please make sure everyone in the classroom has registered for the CPS*3440/5014 course and registered only once!!!
- We have sections CPS*3440*02, CPS*3440*03, and CPS*5014*01 in this classroom.

Today's Topics

- Welcome & Course Kickoff
- Course Information
- Syllabus Highlights
- Why Study Algorithms?

Today's Topics

- **Welcome & Course Kickoff**
- Course Information
- Syllabus Highlights
- Why Study Algorithms?

Instructor

- Dr. Yan Ma (She/Her)
- Ph.D. in CS from Stony Brook University – August 2025
- MS in CS at Stony Brook University – May 2022



Research Interests

- Human-AI Interaction

- Exploring the intersection of Human-Computer Interaction (HCI) and Artificial Intelligence (AI).
 - How do people communicate and collaborate with AI systems?
 - In what ways can AI enhance human experience and performance?
 - How can AI contribute to making technology more inclusive and accessible for diverse user populations?

- Input Technologies

- Text Input

- Accessibility

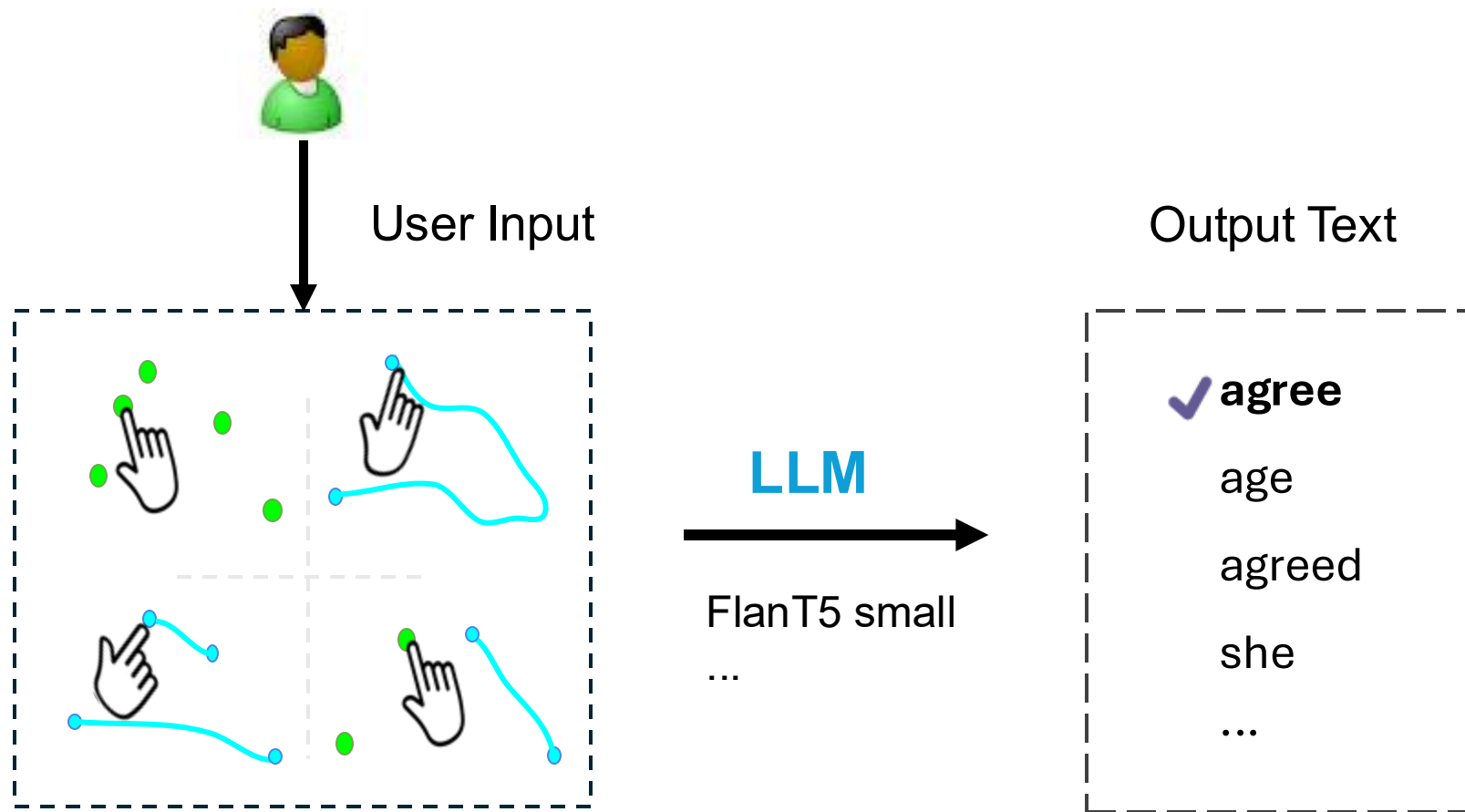
- People with disabilities

- Virtual Reality and Mixed Reality

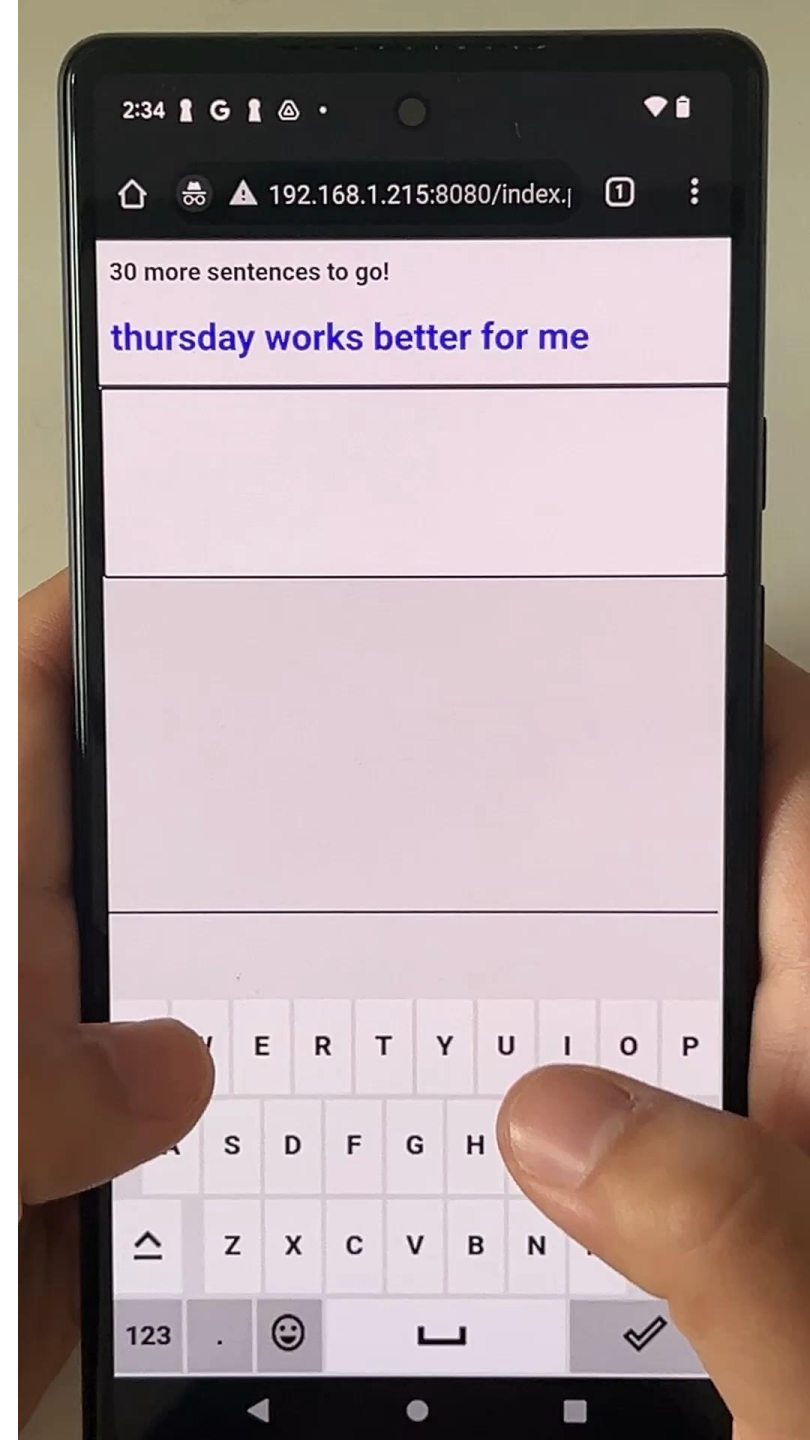
Challenges of Current Keyboard Decoding Algorithms



LLM-Powered Flexible Typing



“Thursday”: tap-gesture
“t-h-u”, “r”, “s”, “d-a-y”



Video speed: 0.5x

“Thursday”: tap-gesture

“t-h-u”, “r”, “s”, “d-a-y”

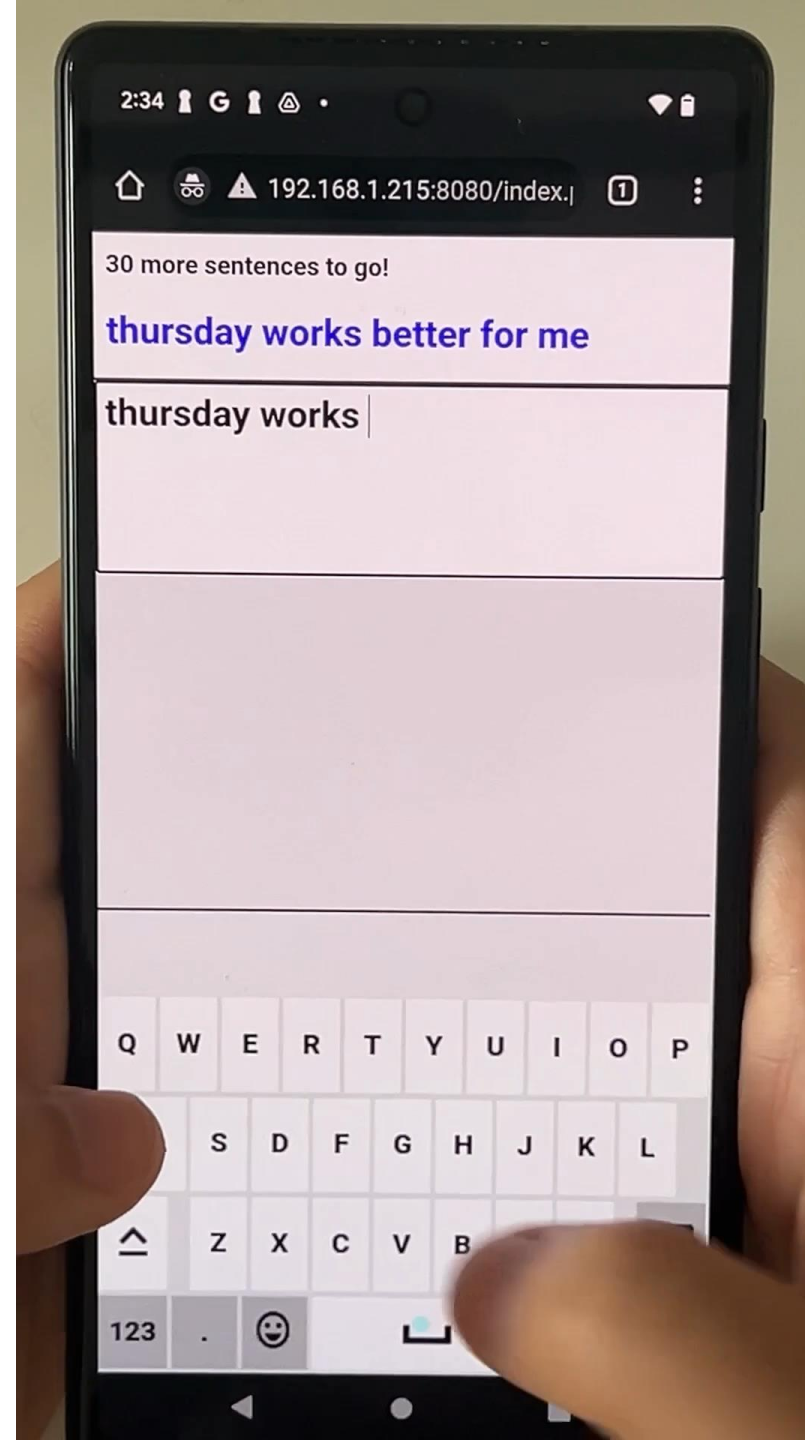
“Works”: tap-gesture

“w”, “o-r-k”, “s”

“Better”: multi-stroke gesture

“b-e”, “t-t-e-r”

Video speed: 0.5x



Research Interests

- Human-AI Interaction

- Exploring the intersection of Human-Computer Interaction (HCI) and Artificial Intelligence (AI).
 - How do people communicate and collaborate with AI systems?
 - In what ways can AI enhance human experience and performance?
 - How can AI contribute to making technology more inclusive and accessible for diverse user populations?

- Input Technologies

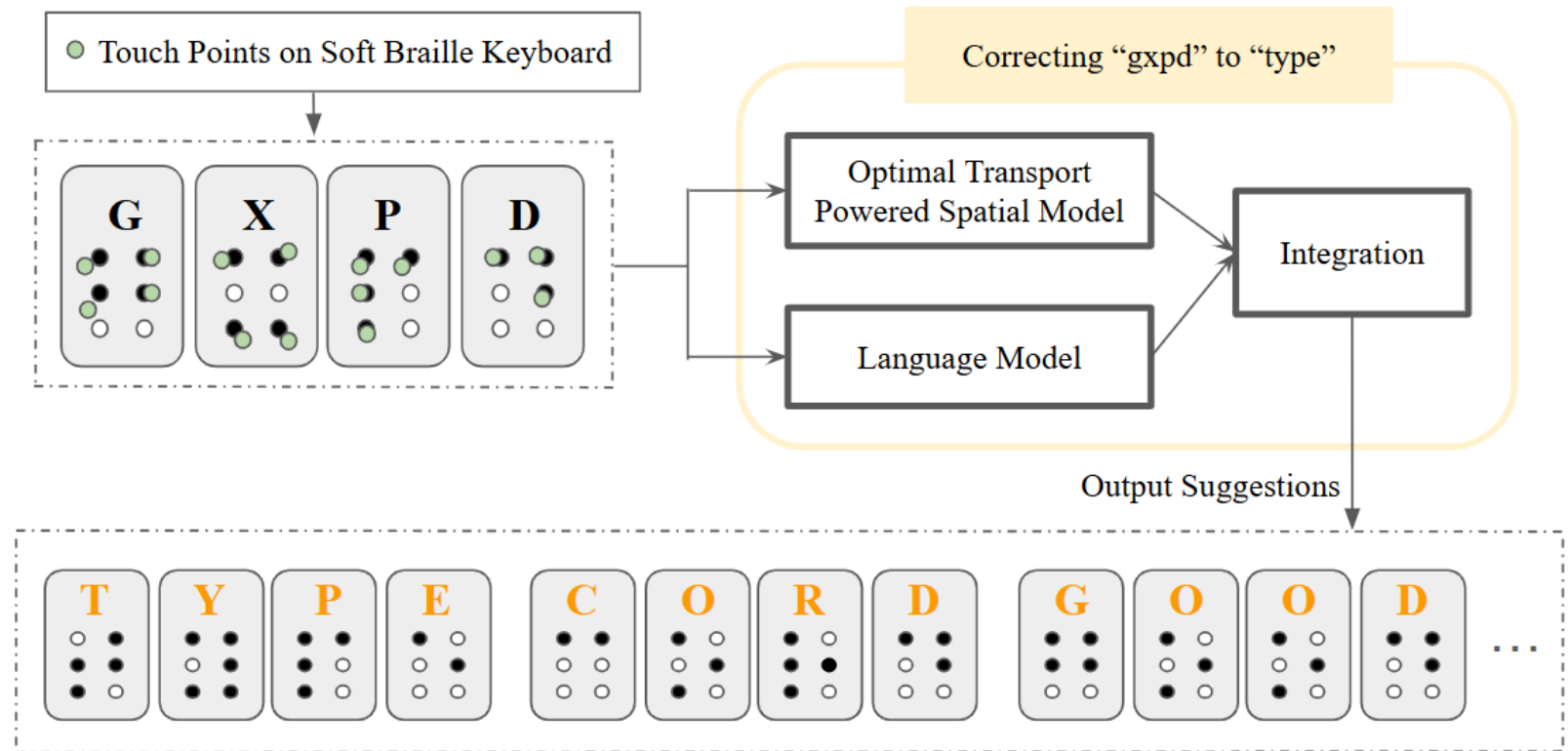
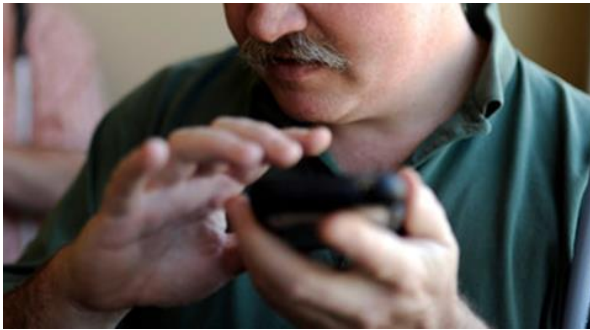
- Text Input

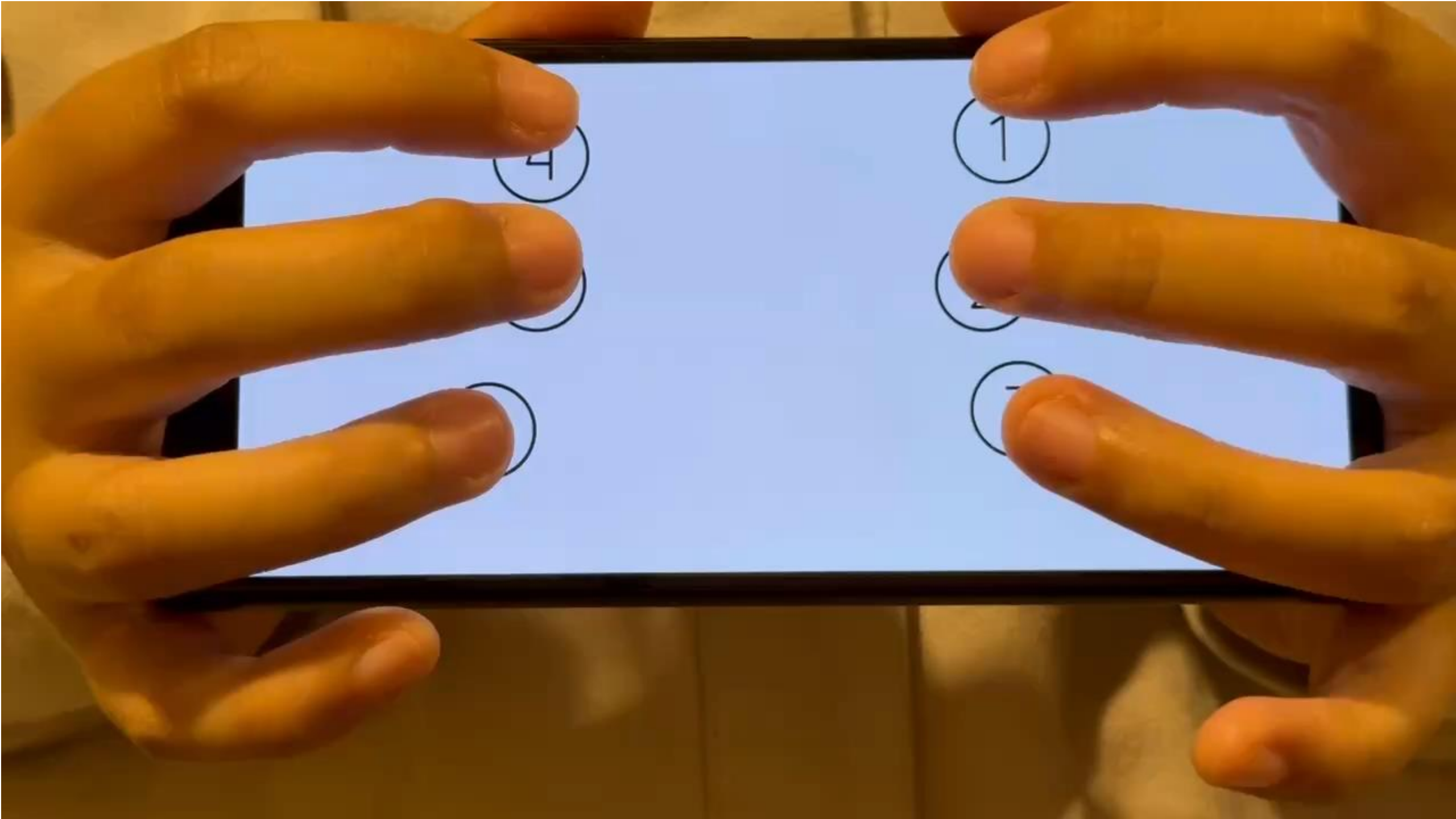
- Accessibility

- People with disabilities

- Virtual Reality and Mixed Reality

Enabling Auto-Correction on Soft Braille Keyboard





Research Interests

- Human-AI Interaction

- Exploring the intersection of Human-Computer Interaction (HCI) and Artificial Intelligence (AI).
 - How do people communicate and collaborate with AI systems?
 - In what ways can AI enhance human experience and performance?
 - How can AI contribute to making technology more inclusive and accessible for diverse user populations?

- Input Technologies

- Text Input

- Accessibility

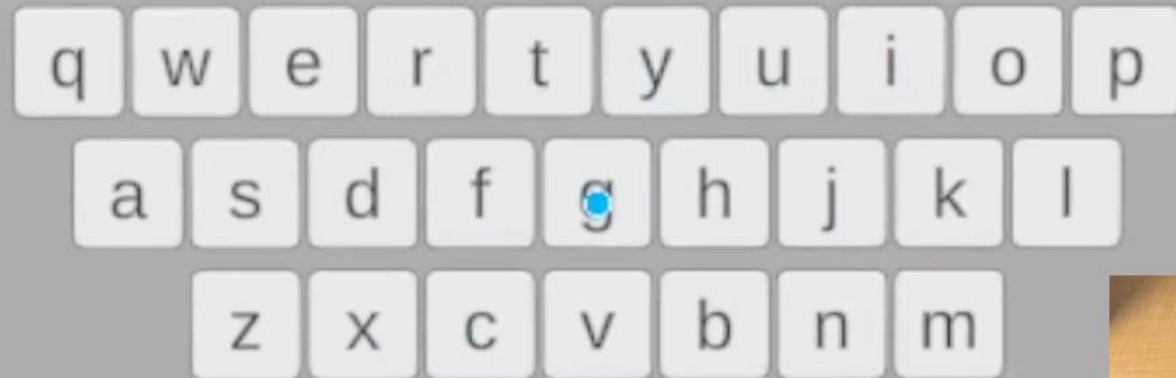
- People with disabilities

- Virtual Reality and Mixed Reality

1/10

communicate through email

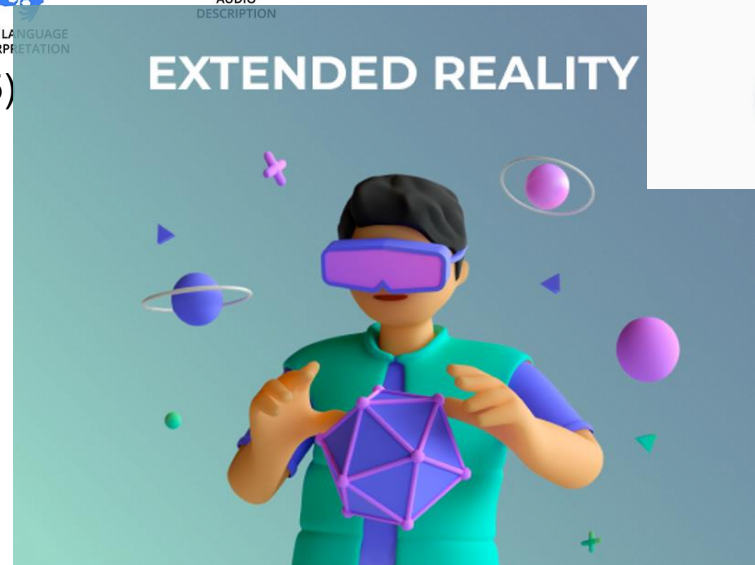
Enter text...



Research Interests



(University of Pittsburgh, 2025)



(Encora, 2022)



(Mobisoft Infotech, 2025)



Let's Get to Know You!

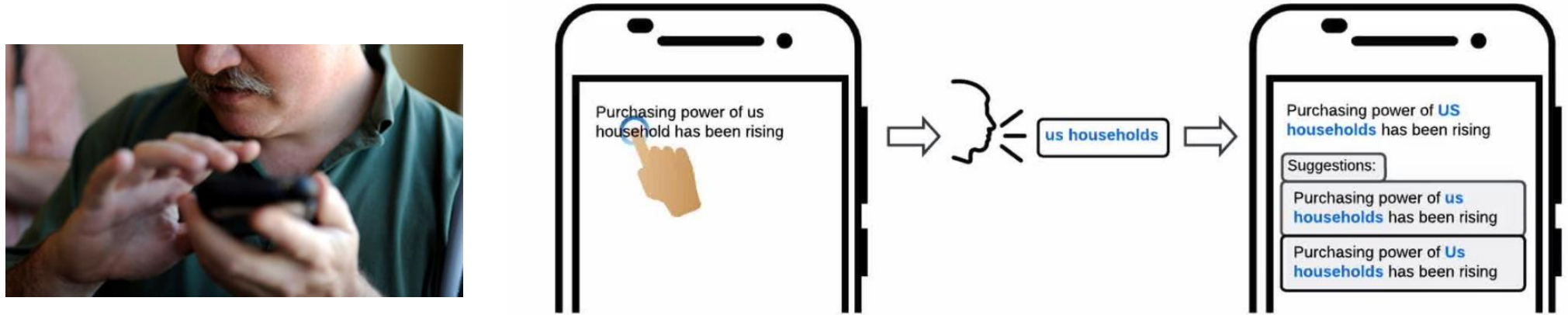
- Your full name
- Preferred nickname, if any
- Programming languages you're familiar with or have used before
- Hobbies or interests (music, sports, gaming, or anything unique you'd like to share!)
- Anything fun or interesting you did over summer break?

Summer experiences/research

- Do you have any research experience before?
- Tell us a little bit about your research

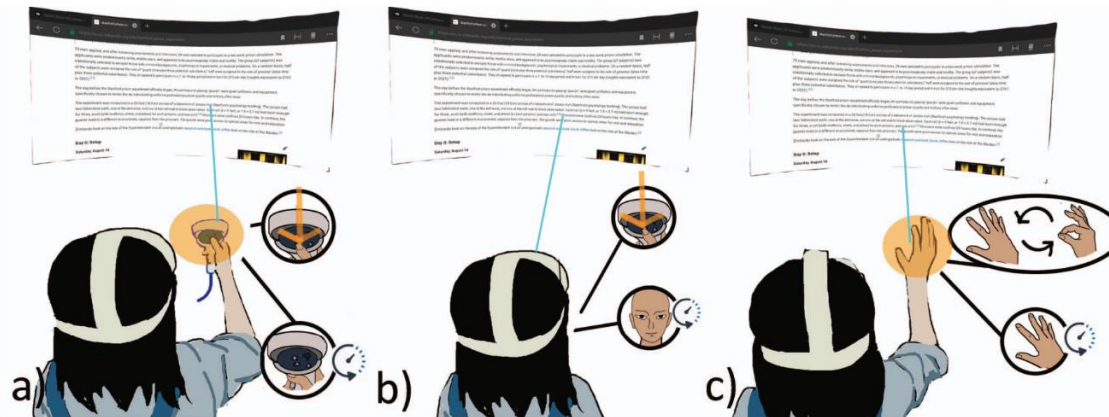
LLM-Powered Text Editing Methods

(1)



(Tap & Say, Zhao et al., CHI 2025)

(2)



(Xu et al., ISMAR 2022)

- **Selection**
- Insertion
- Deletion
- Correction
-
- Copy
- Paste
- Cut
- Highlight
- Bold

Today's Topics

- Welcome & Course Kickoff
- **Course Information**
- Syllabus Highlights
- Why Study Algorithms?

Instructor information

- Dr. Yan Ma
- *Office Location:* Green Lane Academic Building Room 237
- *Office Hours:* TTH 11:00 AM – 12:00 PM
or MW 10:30 AM – 12:00 PM
- *Email:* yama@kean.edu

Communication and Response Expectations

- Feel free to stop by my office during **scheduled office hours**.
- If you prefer to attend virtually, please email me in advance and wait for confirmation before assuming the meeting is confirmed.
- I respond to emails on weekdays within 24–48 hours.
- Unless it's urgent, please **wait at least 24 hours** before sending a follow-up email.
- Extension requests sent within 24 hours of a deadline will not be reviewed.
- I do not respond to emails **after 8:00 PM** on the day an assignment is due.

Contacting Me: Email Guidelines

- When emailing me, please make sure to:
 - Include your **course number and section** in the title
 - State your **full name** clearly
- Write professionally
 - A proper greeting
 - A clear subject line
- **Attach relevant files** if you are asking for help

Contacting Me: An Example Email

Subject: [CPS3440-02] Homework 1 questions

Hi Professor Ma,

I am xxx (your full name) from CPS3440 section 02. I have a question about Q2 in homework 1. My code was compiled successfully without errors, but the output wasn't as expected. Could you please check it for me? I've attached my code to this email. Thanks for your help!

xxx (your preferred name)

Today's Topics

- Welcome & Course Kickoff
- Course Information
- **Syllabus Highlights**
- Why Study Algorithms?

Learning Outcomes

- A. Understand the need for analyzing the efficiency of an algorithm.
- B. Analyze algorithms in order to determine their complexity.
- C. Understand the difference between inefficient algorithms and “hard” problems.
- D. Develop a repertoire of useful algorithms.

Syllabus Overview



Account



Dashboard



Courses



Calendar



Inbox



History



Studio



Commons



Help

 CPS*3440*02.2025FA > Simple Syllabus

Fall 2025



2025FA CPS 3440 02
Last updated Sep 2, 2025

Edit



Home

Announcements 

Simple Syllabus

Syllabus

Modules

Assignments 

Rubrics

Files 

People

Pages

Attendance

Grades

Discussions

Chat

 Student Engagement 1 students viewed

[View Report](#)

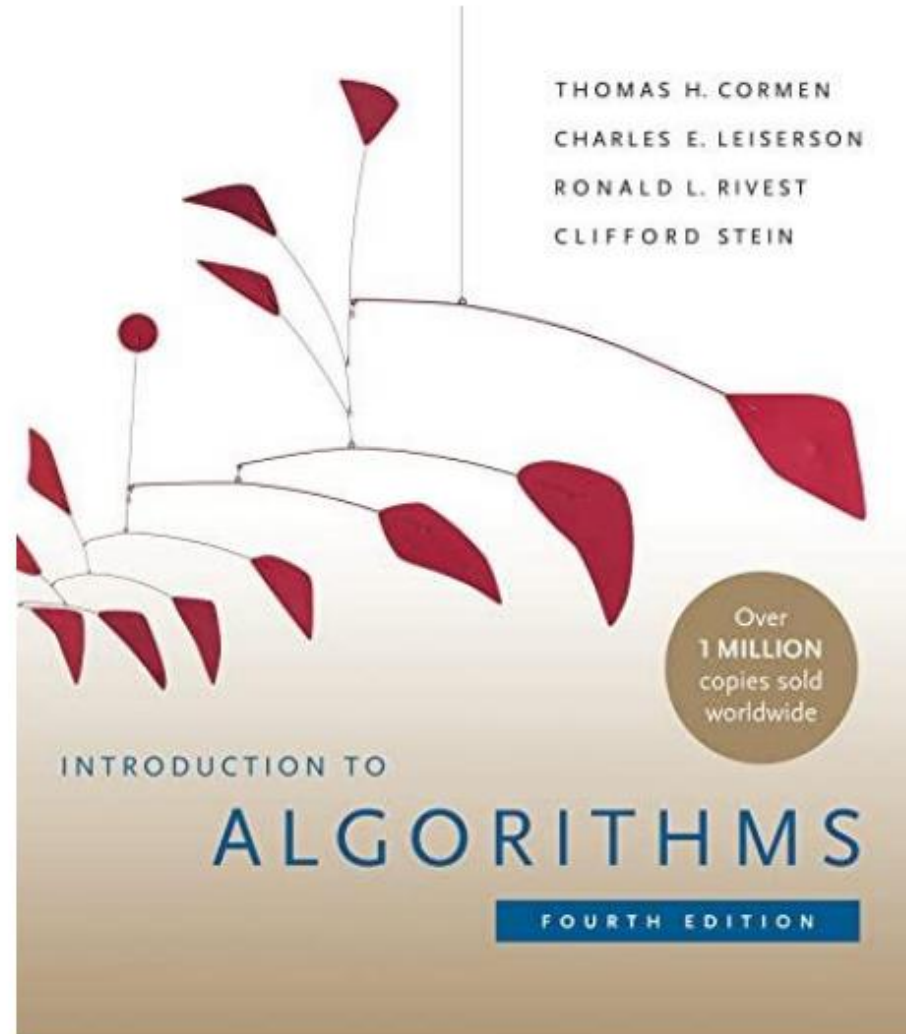


KEAN

Class Information

Course Title: Analysis of Algorithms
Course Number and Section: CPS 3440 02
Campus Location: KEAN UNION
Semester: 2025FA
Class Meeting Days and Times: M W From 14:00 To 15:15
Class Meeting Location: Bldg: GLAB : Room: 307

Textbook



Week/Unit	Topics	Assignments Due
Unit 1 / Course Introduction: The role of algorithms in computing and Foundations review	Textbook Chapters 1, 2, 3, Appendices	Homework 1: 35 Points
Unit 2 / Recursive Functions. Divide-and-Conquer. The Master Theorem.	Chapter 4	Homework 2: 35 Points
Unit 3 / Sorting and Searching Review. Elementary Data Structures Review.	Chapters 10, 12	Homework 3: 35 Points
Unit 4 / Elementary Tree and Graph Algorithms Review	Chapter 20	
Unit 5 / Sorting and Searching: Mergesort, Heapsort, and Quicksort Review	Chapters 3, 6, 7	Homework 4: 35 Points
Unit 6 / Sorting in Linear Time: Counting Sort, Radix Sort, and Bucket Sort	Chapter 8	Homework 5: 35 Points
Unit 7 / Median and Order Statistics	Chapter 9	Exam on Units 1-7: 300 Points

Unit 8 / Dynamic Programming. Properties and Strategy. Analysis. Limitations.	Chapter 14	Homework 6: 35 Points
Unit 9 / Dynamic Programming. Greedy Algorithms. String Matching.	Chapters 15, 32	Homework 7: 35 Points
Unit 10 / Graph Algorithms. Data Structures for Graphs. Topological Sort. Strongly Connected Components.	Chapter 20	
Unit 11 / Minimum Spanning Trees	Chapter 21	Homework 8: 35 Points
Unit 12 / Shortest Paths Algorithms	Chapter 22	Homework 9: 35 Points
Unit 13 / Graph Algorithms: Minimum Cuts	Chapter 24	
Unit 14 / Graph Algorithms: Maximum Flows and Bipartite Matching	Chapters 24, 25	Homework 10: 35 Points
Unit 15 / NP-Completeness	Chapters 34, 35	Exam on Units 8-15: 300 Points
	Attendance	Attendance (in total): 50 points
		Total: 1000 Points

Canvas Course Navigation



Account



Dashboard



Courses



Calendar



Inbox



History



Studio



Help



CPS*3440*02.2025FA > Modules

Fall 2025

Home

Simple Syllabus

Syllabus

Modules

Assignments

People

Pages

Grades

Discussions

Chat

Microsoft OneDrive

Lucid (Whiteboard)

Class Notebook

Collapse All

▼ Ask the Instructor

 Ask the Instructor

▼ Course Information

▼ Unit 1: Introduction

Use the Canvas Calendar to Stay on Track!

The screenshot displays the Canvas LMS interface with the calendar view selected. The sidebar on the left contains navigation links: Account, Dashboard, Courses, Calendar (highlighted), Inbox, History, Studio, Commons, and Help. The main calendar area shows the month of September 2025. The calendar grid has columns for days of the week (SUN to SAT) and rows for weeks. A red box labeled "Homework..." is placed over the date 15. The right-hand panel shows the "CALENDARS" section with a list of calendars, including "Yan Ma" and several course-related calendars. The "OTHER CALENDARS" section is also visible at the bottom right.

Today ← → September 2025 Week Month Agenda +

SUN	MON	TUE	WED	THU	FRI	SAT
31	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	1	2	3	4

▼ CALENDARS

- ☐ Yan Ma
- ☐ 2025FA*CPS*2231*06 (Computer Programming)
- ☐ 2025FA*CPS*2231*07 (Computer Programming)
- ☒ 2025FA*CPS*3440*02 (Analysis of Algorithms)
- ☐ 2025FA*CPS*3440*03 (Analysis of Algorithms)
- ☐ 2025FA*CPS*5014*01 (Gr Analysis of Algorithms)
- ☐ New Faculty Onboarding Course

▼ OTHER CALENDARS NEW +

Grading Policy

The course includes two exams, 10 homework assignments, and attendance.

Breakdown:

- Attendance: 5%
- Homework Assignments (10): 35%
- Midterm Exam: 30%
- Final Exam: 30%

Late policy

- Submitted within 24 hours after the deadline: 10% deduction from the earned grade
- Submitted between 24 and 48 hours after the deadline: 40% deduction from the earned grade
- **Submitted more than 48 hours after the deadline: Will not be accepted or graded**
- No exceptions will be made unless there are serious circumstances supported by proper documentation.

Assignment and Exam Format

- **Homework Assignments:**

Submit your work as a PDF, Word (.docx), or a clear photo of your handwritten solution. You may complete assignments on paper, in Word/Google Docs, or using LaTeX.

- **Exams:**

All exams will be conducted **in class, closed book**. You are allowed **one cheat sheet** (standard letter size) unless otherwise specified.

- **Attendance:**

To be marked present, please write your full name on the sign-in sheet at the beginning of each class.

Makeup Exam Policy

1. Makeup exams will be allowed only in cases of **medical or family emergencies**, supported by **official documentation** such as a **doctor's note, hospital records, or police reports**.
2. Makeup exams must be completed **within 7 days of the original exam date** unless the documentation specifies a longer extension.

Class policies

- Extra points opportunities will be announced throughout the semester.
- A grade of **C or better** is required for CS major.
- Grades can be checked online using Canvas (refer to Section My Grades)
- Students are responsible for reporting any grade errors to the instructor.
- Provide accommodation documentation early in the semester (extended exam time, .etc, will be provided only based on it)

Academic Honesty and AI Policy

- **No Plagiarism**. All work turned in under your name must be your own work.
- No credit will be given for an assignment that is copied in part or in total from another person. If plagiarism happens twice, you will fail the class.
- Use of ChatGPT or any other AI tools for the midterm or final exam will result in a failing grade for the course.
- Use of ChatGPT or any other AI tools for assignments will result in a grade of zero for that assignment.

Attendance policy

- Starting with the third absence, documentation is needed to waive the penalty, like a doctor's note, a photo of a non-functional car, or a letter from an employer stating work obligations during lecture hours.
- An email like: “Hey professor, I can’t make it for today’s class because I am sick” without any doctors’ documents will work for the first two absences, but for the third one, it does not work.
- Students must email the instructor to justify each absence. For the initial two absences, no evidence is required to avoid a penalty.

Important dates

- Mark the important dates on your calendar:
- Last day to add courses: Tue., Sep. 9, 2025
- Last day to drop without “W” mark (100% refund): Tue., Sep. 9, 2025
- Last day to withdraw with “W” grade (75% refund): Tue., Sep. 16, 2025
- Last day to withdraw with “W” grade (50% refund): Tue., Sep. 23, 2025
- Last day to withdraw with “W” grade (no refund): Mon., Nov. 17, 2025
- See the Academic Calendar and **Double-check—adhere to Calendar**

<https://www.kean.edu/offices/registrars-office/academic-calendar>



Questions/Concerns?

Today's Topics

- Welcome & Course Kickoff
- Course Information
- Syllabus Highlights
- **Why Study Algorithms?**

Why Study Algorithms?

Their impact is broad and far-reaching.

Internet. Web search, packet routing, distributed file sharing, ...

Biology. Human genome project, protein folding, ...

Computers. Circuit layout, file system, compilers, ...

Computer graphics. Movies, video games, virtual reality, ...

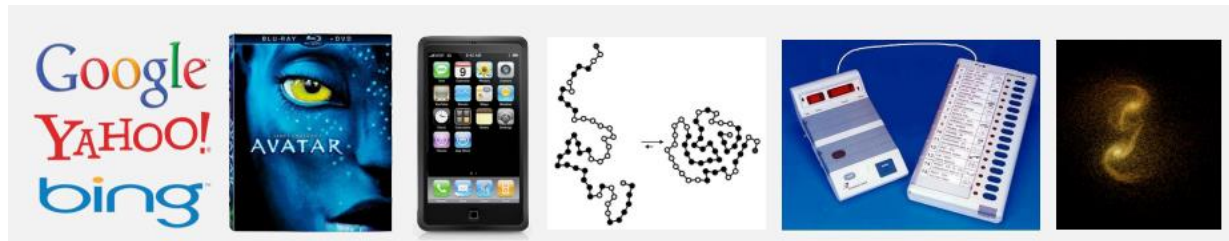
Security. Cell phones, e-commerce, voting machines, ...

Multimedia. MP3, JPG, DivX, HDTV, face recognition, ...

Social networks. Recommendations, news feeds, advertisements, ...

Physics. N-body simulation, particle collision simulation, ...

...



Why Study Algorithms?

Old roots, new opportunities.

- Study of algorithms dates at least to Euclid.
- Formalized by Church and Turing in 1930s.
- Some important algorithms were discovered by undergraduates in a course like this!



Why Study Algorithms?

For intellectual stimulation.

“ For me, great algorithms are the poetry of computation. Just like verse, they can be terse, allusive, dense, and even mysterious. But once unlocked, they cast a brilliant new light on some aspect of computing. ” — Francis Sullivan

“ An algorithm must be seen to be believed. ” — Donald Knuth



Why Study Algorithms?

To become a proficient programmer.

“I will, in fact, claim that the difference between a bad programmer and a good one is whether he considers his code or his data structures more important. Bad programmers worry about the code. Good programmers worry about data structures and their relationships.”

— *Linus Torvalds (creator of Linux)*



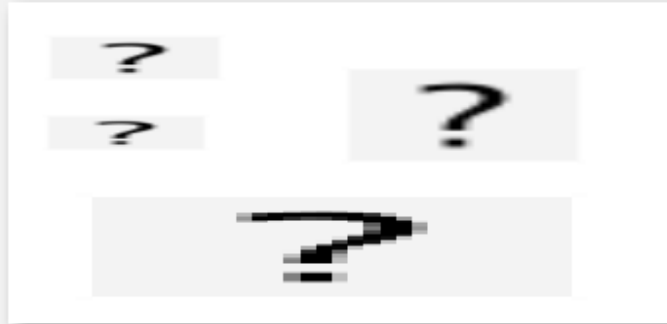
“Algorithms + Data Structures = Programs.” — Niklaus Wirth



Why Study Algorithms?

They may unlock the secrets of life and of the universe.

Scientists are replacing mathematical models with computational models.



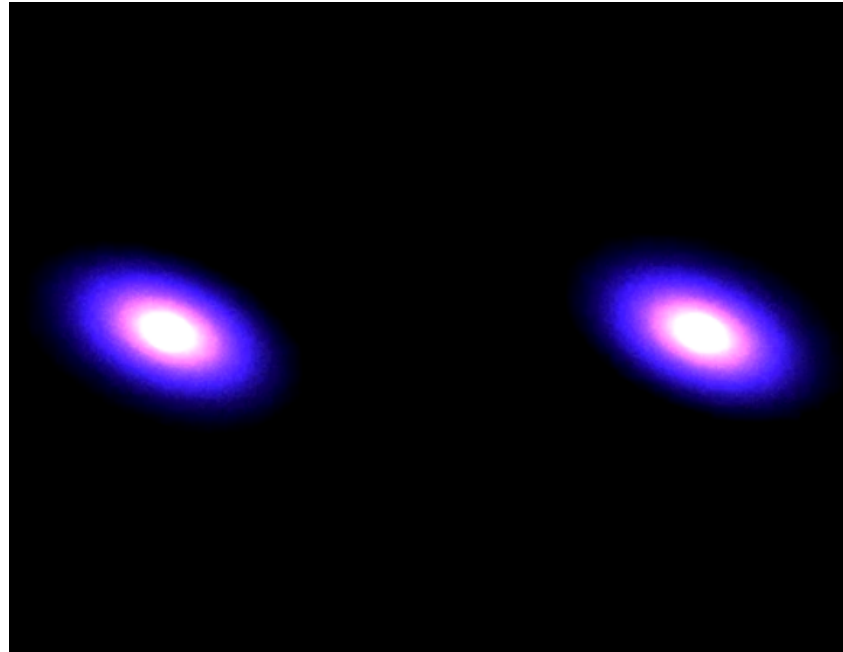
20th century science
(formula based)

```
for (double t = 0.0; true; t = t + dt)
  for (int i = 0; i < N; i++)
  {
    bodies[i].resetForce();
    for (int j = 0; j < N; j++)
      if (i != j)
        bodies[i].addForce(bodies[j]);
  }
```

21st century science
(algorithm based)

Why Study Algorithms?

To solve problems that could not otherwise be addressed.



$O(N \log N)$ parallel N-body simulation using 32 GPUs

Why Study Algorithms?

For fun and profit.



Today's Topics

- Welcome & Course Kickoff
- Course Information
- Syllabus Highlights
- Why Study Algorithms?